8-1 Computer Networks

Point!

1 Information and Communication Networks

(1) ([1]**Information and communication network**) : A network resembling a web that utilizes communication lines to transmit information.

(2) ([2]**Computer network**) : A communication system that connects information communication devices like computers and smartphones, enabling them to exchange data with one another.

(3) Types of Networks

[1] ([3]**LAN**) : A network that connects devices within a limited area, such as within a school or home.

- ([4]**Wired LAN**) : A method of exchanging data by directly connecting to cables.
- ([5]**Wireless LAN**) : A method of exchanging data via wireless radio waves instead of cables.

[2] ([6]**WAN**) : A network that connects a broader area.

(4) ([7]**Internet**) : A network that connects LANs and WANs on a global scale.

(5) Connecting to the Internet

[1] ([8]**ISP (provider)**) : A company that facilitates connection to the internet. It is necessary to sign a contract to connect to the internet.

[2] ([9]**Router**) : A device that relays and forwards data between different networks.

[3] ([10]**Hub**) : A central device that connects multiple cables.

2 Network Usage Patterns

(1) Server and Client

[1] ([11]**Server**) : A computer that provides various services.

- ([12]**File server**) : A server that manages files.
- ([13]**Printer server**) : A server that handles print processing.
- ([14]**Mail server**) : A server that handles the sending and receiving of emails.
- ([15]**Proxy server**) : A server that acts as an intermediary for accessing the internet.

[2] ([16]**Client**) : A computer that requests various services from a server.

(2) Network Usage Patterns

[1] ([17]**Client-server system**) : A method where services are provided and received between a server and a client.

[2] ([18]**Peer-to-peer system**) : A method where computers are in an equal relationship, providing and receiving services to and from each other.

3 Communication Methods

(1) ([19]**Circuit switching method**) : A communication method where a communication path is established between the sender and the recipient before starting the transmission to exchange information.

(2) ([20]**Packet**) : A unit for dividing data into small chunks when transmitting data over a network.

(3) ([21]**Packet switching method**) : A communication method that involves dividing the data being transmitted into packets for exchange.

(4) Advantages and Disadvantages of Communication Methods

	Circuit switching method	Packet switching method
Advantages	Once a communication path is established, stable communication can be achieved.	Even if the network is congested, data can be sent gradually, so the possibility of being completely unable to communicate is low.
Disadvantages	The communication path is occupied by the user, so it is not possible to use the line simultaneously with others.	Packets can sometimes be lost along the way or be affected by network congestion.

Warm Up

Answer the following questions.

(1) Choose one statement that is incorrect regarding network construction devices from the options **A** to **D** below, and answer using the letter.

A Organizations or companies that facilitate connections to the internet are referred to as providers.

B A network limited to areas such as within a school or a home is referred to as a WAN.

C A device that gathers and connects multiple cables is referred to as a hub.

D A device that relays and transfers between different networks is referred to as a router.

(2) Choose the statement that best fits into the blanks [1] to [4] in the following table from the options **A** to **D** below, and answer using the letters.

	Advantages	Disadvantages
Circuit switching method	[1]	[2]
Packet switching method	[3]	[4]

A The line is occupied, which ensures stable communication.

B There may be delays or loss of data during transmission.

C Only the number of lines necessary for communication is required.

D It is possible to handle many communications simultaneously.

Explanation

(1) A network limited to areas such as within a school or a home is referred to as a LAN. Therefore, **B**

(2) [1] **A** [2] **C** [3] **D** [4] **B**

Try

1 Choose the term that best fits into the blanks [1] to [6] in the following sentences from the options **A** to **H** below, and answer using the letters.

> A network within a relatively limited area, such as within a school or home, is referred to as a ([1]), while a network that covers a wider area is referred to as a ([2]). A network that connects ([1]) and ([2]) on a global scale is called the ([3]). In order for regular households to use it, they need to have a contract with a business that facilitates connection to the internet, known as a ([4]).
>
> Additionally, computers within a local range are typically connected via cables through a concentrator device known as a ([5]). When connecting different networks, an intermediary device called a ([6]) is used.

A LAN **B** WAN **C** Wi-Fi **D** Hub
E Server **F** Router **G** Internet **H** ISP (provider)

2 Answer the following questions.

(1) Choose the term that best fits into the blank in the following sentence from the options **A** to **D** below, and answer using the letter.

> A system configuration in which computers connected through a network are structured to allocate roles as service providers and service receivers is referred to as ( ).

A Client-server system **B** Peer-to-peer system
C Local Area Network (LAN) **D** Wide Area Network (WAN)

(2) Fill in the blanks with the appropriate words.

[1] A server that acts as an intermediary for accessing the internet is referred to as a ( ).
[2] A server that manages files is referred to as a ( ).
[3] A computer that requests various services from a server is referred to as a ( ).

3 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> There are ([1]) methods and ([2]) methods in communication systems. The ([1]) method is a communication method where a communication path is established between the sender and the recipient before starting communication to exchange information. The ([2]) method is a communication method where the data to be sent is divided into small units called packets for exchange.

(2) Choose all appropriate statements that describe circuit switching method and packet switching method from the options **A** to **D** below, and answer using the letters.

A Circuit switching method allows for high communication speeds because the communication path is shared with other users.
B In circuit switching method, once a connection is established, stable communication is possible because the communication path is occupied.
C Packet switching method involves sharing the communication path, which can result in lower communication speeds.
D Packet switching method has a low likelihood of complete communication failure, even when the network is congested.

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of a WAN from the options **A** to **D** below, and answer using the letter.

A It can only be connected within limited buildings such as schools, companies, and homes.

B It is possible to connect areas that are far apart.

C It is a communication network that connects information and communication devices, such as computers, to exchange information with each other.

D It is possible to interconnect networks around the world.

(2) Choose the term that best fits into the blanks in the following sentences from the options **A** to **D** below, and answer using the letters.

[1] The method of exchanging data by directly connecting cables is referred to as ( ).

[2] The method of exchanging data using wireless radio waves instead of cables is referred to as ( ).

[3] The device that relays and forwards data between different networks is referred to as a ( ).

A Wired LAN **B** Router **C** Hub **D** Wireless LAN

2 Answer the following questions.

(1) Choose one statement that is incorrect regarding the usage patterns of networks from the options **A** to **D** below, and answer using the letter.

A A server that manages data and files on a network is referred to as a file server.

B A computer that requests various services from a server is referred to as an ISP (provider).

C A server that accesses the internet on behalf of a user is referred to as a proxy server.

D A method where computers are in an equal relationship, providing and receiving services to and from each other, is referred to as a peer-to-peer system.

(2) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below, and answer using the letters.

- A server that handles printing processes such as copying and printing is referred to as a ([1]).
- A server that facilitates the sending and receiving of emails between computers is referred to as a ([2]).
- The usage pattern where a server and a client connect over a network and exchange services is referred to as ([3]).

A Client-server system **B** Mail server

C Internet **D** Printer server

(3) If the following statements [1] to [4] pertain to circuit switching method, answer with “A.” If the statements pertain to packet switching method, answer with “B.”

[1] Once a communication path is established, stable communication can be achieved.

[2] Since it can send data in small portions, the possibility of being unable to communicate is low, even if the network gets congested.

[3] Delays, tampering, or loss may occur during data transmission.

[4] Since you cannot use the line simultaneously with others, there is a need for a number of lines dedicated to communication.

8-2 IP Addresses and Domain Names

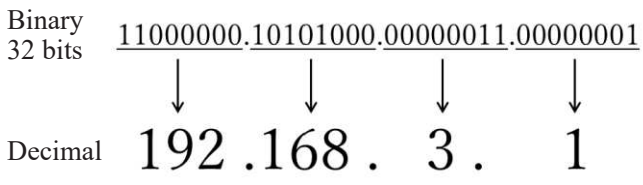
Point!

IP Addresses and Domain Names

(1) ([1]**IP address**) : A unique identification number equivalent to an address assigned to devices on an information and communication network.

<Features of IP Addresses>

- The method of representing an IP address using binary with ([2]**32 bits**) is referred to as ([3]**IPv4**).
- It is represented in four blocks of ([4]**8 bits**) each, separated by a dot (.) and expressed in decimal from 0 to ([5]**255**).



(2) Global IP Address and Private IP Address

[1] ([6]**Global IP address**) : An IP address used on the internet.

[2] ([7]**Private IP address**) : An IP address used within a local network, such as a LAN.

(3) Issue of IP Address Exhaustion

Due to the rapid proliferation of the internet, there are almost no new IPv4 addresses available for allocation. Therefore, we are currently transitioning to ([8]**IPv6**) IP addresses, which have been expanded to ([9]**128 bits**).

(4) ([10]**Domain name**) : A string of characters assigned to make a numerical IP address more understandable for humans.

(5) ([11]**DNS**) : A system that associates domain names with IP addresses and vice versa. The ([12]**DNS server**) is responsible for fulfilling that role.

Warm Up

Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

A 192.168.0.1
B kantei.go.jp
C 555-1234-5678
D https://www.example.co.jp/

Explanation

An IPv4 address is divided into four blocks, with each block's value ranging from 0 to 255.
Therefore, **A**

Try

Answer the following questions.

(1) Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

A 00.00.11.aa.bb.cc
B 050-1234-5678
C 10.123.45.67
D https://www.example.co.jp/

(2) Fill in the blank in the following sentence with the correct term.

An IP address that is uniquely assigned on the internet with no duplication is referred to as a ( ) IP address.

(3) Choose the correct statement regarding IP addresses from the options **A** to **D**, and answer using the letter.

A The number of IP addresses in a four-byte format like 192.168.1.1 exceeds the population of the Earth, which is approximately 7 billion.
B The issue of IP address exhaustion is resolved by IPv6, which is represented using 128 bits.
C In IPv4, addresses are represented by combining four numbers ranging from 0 to 256.
D A global IP address can be freely used within a local network, such as within the same company.

Exercise

1 Answer the following questions.

(1) Choose the appropriate example of an IPv4 IP address for a computer from the options **A** to **D** below, and answer using the letter.

A saiyo@example.co.jp
B 070-1234-5678
C https://www.example.co.jp/
D 203.216.206.63

(2) Fill in the blank in the following sentence with the correct term.

An IP address that can be freely used within a local network, such as a LAN, is referred to as a ( ) IP address.

(3) Choose the correct explanation regarding DNS from the options **A** to **D** below, and answer using the letter.

A A unique identification number assigned to devices on an information and communication network.
B A string of characters assigned to an IP address to make it easier for humans to understand.
C A system that maps domain names to IP addresses and vice versa.
D An IP address that is uniquely assigned on the internet with no duplication.

8-3 Communication Protocol

Point!

1 Communication Protocol

(1) ([1]**Packet**) : The unit used when data is divided into small chunks to be transmitted over a network. Sent with a ([2]**header**) that includes the destination information.

(2) ([3]**Communication protocol (protocol)**) : A common agreement in information and communication networks.

[1] ([4]**TCP**) : A protocol that divides the data to be sent into packets, arranges the received packets in order, and requests the retransmission of any packets that were lost during communication.

[2] ([5]**UDP**) : A protocol that emphasizes sending data in real time. Used for voice calls and streaming video.

[3] ([6]**IP**) : The protocol that assigns IP addresses to deliver packets to the correct destination.

2 The Mechanism of Communication in TCP/IP

(1) ([7]**TCP/IP**) : A set of protocols used on the internet. The sending and receiving of data is controlled across four layers.

<The 4-Layer Model of TCP/IP>

Layer	Layer names	Main functions	Examples of protocols
Four layers	([8]**Application layer**)	Communication between applications	HTTP, SMTP, etc.
Three layers	([9]**Transport layer**)	Communication control, error detection, and data retransmission	TCP, UDP, etc.
Two layers	([10]**Internet layer**)	IP address allocation and routing decision	IP, etc.
One layer	([11]**Network interface layer**)	Physical connection, interaction between devices	Ethernet, etc.

(2) Mechanism of Data Communication on the internet

1) Divide the data you want to send into ([12]**packets**).

2) Attach the sender or recipient ([13]**IP addresses**) to the header of each packet.

3) Select the optimal communication route for each packet (this is called ([14]**routing**)) and deliver it to the destination.

4) If there any packets are missing upon receipt, a retransmission is requested. The packets are then rearranged in order to complete the data in its entirety.

Warm Up

Answer the following questions.

(1) For the following statements **A** to **D** concerning protocols, mark "○" if the statement is appropriate, and "×" if it is incorrect.

A A communication protocol refers to enhancing security by encrypting data.

B UDP is a protocol that ensures reliable communication by guaranteeing data integrity and order.

C TCP is a protocol that prioritizes data transfer speed, and it ensures the integrity and order of data while minimizing latency.

D IP is a protocol used to assign IP addresses to deliver packets to the correct destination.

(2) Data communication on the internet is transmitted according to TCP/IP. Rearrange the following steps **A** to **D** in the order of communication on the internet.

A Assign IP address of the recipient or the sender.

B Split the data into packets.

C Select the destination and route for the data and deliver it to its destination.

D Reorder the packets in sequence.

Explanation

(1) **A** A communication protocol is a common agreement in information and communication networks. Therefore, ×

B UDP is a protocol that emphasizes sending data in real-time. To prioritize data transfer speed, it does not guarantee the integrity or order of the data. Therefore, ×

C TCP is a protocol that divides the data to be sent into packets and retransmits any packets that are lost during transmission, ensuring reliable communication. Therefore, ×

D ○

(2) **B → A → C → D**

Try

1 Choose the most appropriate explanation of a communication protocol from the options **A** to **D** below, and answer using the letter.

A Cable hub used for connecting devices such as computers and printers to a LAN
B Description of the location of information specified in a web browser and how to retrieve it
C The number used to identify a computer in internet communication
D Protocols for communication over a network

2 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [7] in the following sentences from the options **A** to **H** below, and answer using the letters.

> The rules and procedures for communication over an information and communication network are called ([1]). On the internet, ([2]) is primarily used. Among them, ([3]) is responsible for dividing the data to be sent into ([4]), rearranging the received ([4]) in order, and retransmitting any data that may be missing during communication. On the other hand, ([5]) is responsible for attaching the ([6]), which is the destination address, to ([4]) and sending them to the destination. At this time, ([4]) are transmitted via the internet, searching for the most efficient route to the recipient computer. This is referred to as ([7]).

A TCP/IP **B** IP **C** TCP **D** Packets
E IP address **F** MAC address **G** Routing **H** Communication protocol

(2) The layers in TCP/IP for data communication over the internet are summarized in the following table. Choose the term that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

Layer	Layer names	Main functions
Four layers	([1])	Communication between applications
Three layers	([2])	Communication control, error detection, and data retransmission
Two layers	([3])	IP address allocation, routing decision
One layer	([4])	Physical connection, interaction between devices

A Transport layer **B** Network interface layer
C Internet layer **D** Application layer

(3) Choose the appropriate example for each of the protocols [1] to [4] from the options **A** to **D** below, and answer using the letters.

[1] Transport layer [2] Network interface layer
[3] Internet layer [4] Application layer

A UDP **B** IP **C** SMTP **D** Ethernet

Exercise

1 Choose the term that best fits into the blanks in the following sentences from the options **A** to **D** below, and answer using the letters.

(1) A unique identification number equivalent to an address assigned to devices on an information and communication network is referred to as an ( ).

(2) A common set of rules within an information and communication network is referred to as a ( ).

A IP address **B** TCP/IP **C** Protocol **D** Route

2 Answer the following questions.

(1) Choose the most closely related options for the following sentences from the options **A** to **D** below, and answer using the letters.

[1] A protocol that emphasizes sending data in real time. Used for voice calls and streaming video.

[2] Data that is divided when communicating over a network.

[3] Protocols used on the internet. Divided into four layers.

[4] A protocol that divides the data to be sent into packets and retransmits any packets that are lost during transmission.

A TCP/IP **B** TCP **C** Packet **D** UDP

(2) The layers in TCP/IP for data communication over the internet are summarized in the following table. Choose the function that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

Layer	Layer names	Main functions	Examples of protocols
Four layers	Application layer	([1])	HTTP, SMTP, etc.
Three layers	Transport layer	([2])	TCP, UDP, etc.
Two layers	Internet layer	([3])	IP, etc.
One layer	Network interface layer	([4])	Ethernet, etc.

A Physical connection, interaction between devices
B Communication between applications
C IP address allocation, routing decision
D Communication control, error detection, and data retransmission

(3) The following statements describe the mechanism of data communication on the internet. Answer the following questions.

1) Divide the data you want to send into (**A**).
2) Assign the (**B**) of the sender or recipient to the header of each packet.
3) Select the optimal communication route for each packet and deliver it to its destination.
4) If there any packets are missing upon receipt, a retransmission is requested. The packets are then rearranged in order to complete the data in its entirety.

[1] Complete the sentences by filling in the blanks **A** and **B** with the appropriate terms.

[2] Answer with the appropriate term for the underlined section.

8-4 Mechanism of Web Pages and Emails

Point!

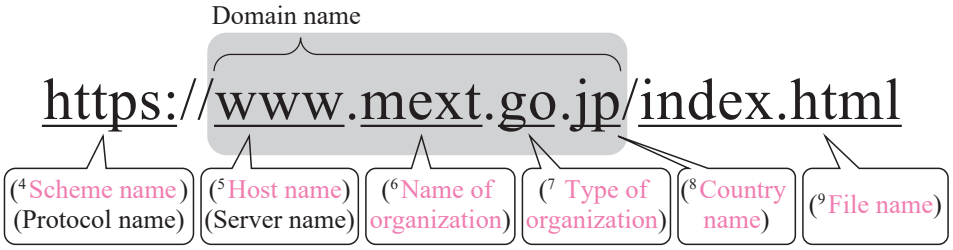
1 Mechanism of the Internet

(1) ([1]**WWW**) : A system that links countless pieces of content on the internet. Users can browse web pages that compile various data. It is commonly referred to as the Web.

(2) ([2]**HTML**) : A language used for creating web pages.

(3) ([3]**URL**) : Represents the location of a web page and the method of communication.

<Structure of a URL>



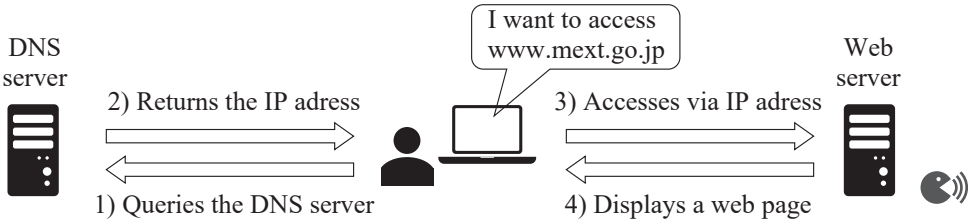
<Examples of Types of Organizations>

Type	Meaning
gov	Government agency
co	Corporation
edu	School (elementary, junior high, and high schools)
ac	University/educational corporation

<Examples of Country Codes>

Code	Country name
eg	Egypt
jp	Japan
hk	Hong Kong
kr	South Korea

(4) Process of Browsing Web Pages



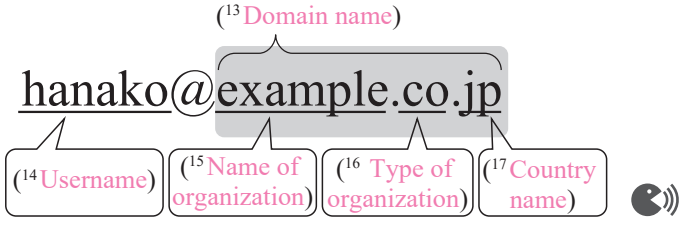
2 Mechanism of Email

(1) Email : A system for exchanging messages and data with specific users via the internet. When sending an email, you use an ([10]**email address**).

(2) Structure of an Email Address

[1] ([11]**Username**) : Identifies the recipient.

[2] ([12]**Domain name**) : Identifies the mail server that the recipient is using.

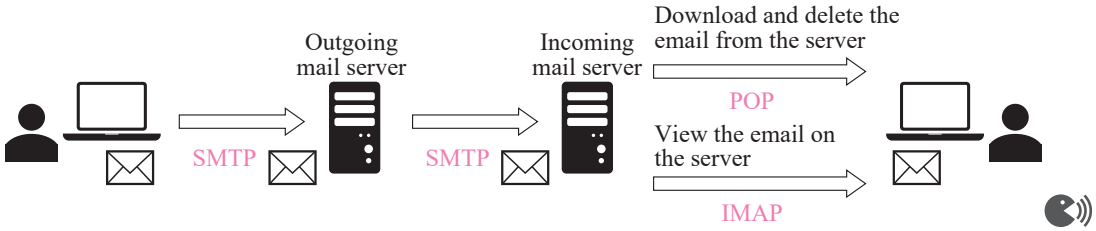


(2) Process of Sending and Receiving Emails

[1] ([18]**SMTP**) : A protocol for sending emails to a mail server.

[2] ([19]**POP**) : A protocol used to download emails from a server and delete them from a server.

[3] ([20]**IMAP**) : A protocol for accessing emails stored on a mail server.



Warm Up

Answer the following questions.

(1) You entered "https://www.example.co.jp" in the web browser and displayed the web page. Rearrange the steps **A** to **D** in the order of processing to display the web page.

A The web server sends the data of the web page.

B Access the IP address returned from the DNS server.

C The DNS server returns the IP address of the web server corresponding to the domain name.

D Query the DNS server for the IP address corresponding to the domain name.

(2) Fill in the blanks [1] to [4] in the following sentence with the appropriate terms.

> An email address is composed of "username@([1])," and the protocol ([2]) is used for sending emails, while ([3]) or ([4]) is utilized for receiving emails.

Explanation

(1) **D → C → B → A**

(2) [1] Domain name [2] SMTP [3] POP [4] IMAP (note: [3] and [4] can be in any order)

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [4] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Web pages are written in a language called ([1]) and the location and communication protocol of the web page are specified by ([2]). The system for converting domain names and IP addresses within ([2]) is ([3]), and ([4]) fulfills that role.

A URL **B** DNS **C** HTML
D WWW **E** DNS server

(2) What is the name of the service that allows you to view web pages on the internet?

(3) Choose the appropriate names for sections [1] to [6] that compose the following URL from the options **A** to **G** below, and answer using the letters.

https://www.mext.go.jp/index.html
[1] [2] [3] [4] [5] [6]

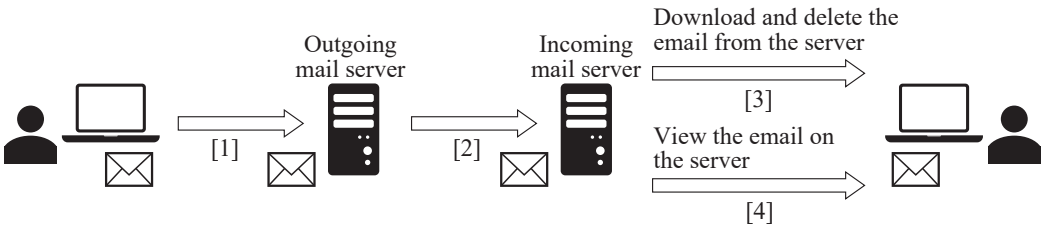
A Scheme name **B** Country name **C** Organization type name **D** Name of organization
E Hostname **F** File name **G** Domain name

2 Answer the following questions.

(1) Name the components labeled [1] and [2] that constitute the following email address.

hanako@example.ed.jp
[1] [2]

(2) The following diagram represents the flow of sending and receiving emails. Choose the appropriate combination of protocols that fit in [1] to [4] in the following diagram from the options **A** to **D** below.



	[1]	[2]	[3]	[4]
A	POP	SMTP	SMTP	IMAP
B	POP	POP	IMAP	SMTP
C	SMTP	SMTP	POP	IMAP
D	SMTP	SMTP	IMAP	POP

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of a DNS from the options **A** to **D** below, and answer using the letter.

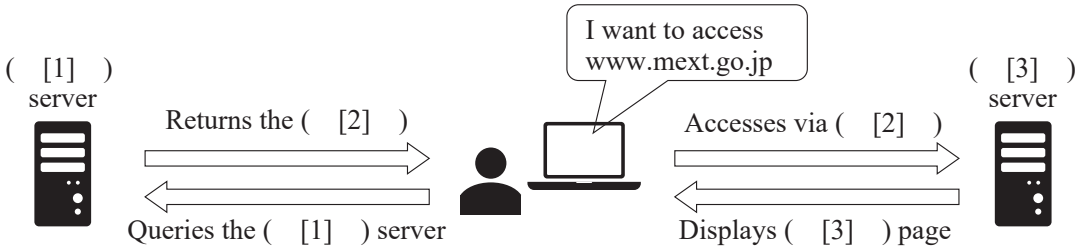
A A system that links countless pieces of content on the internet and is commonly called “the Web.”

B Represents the location of a web page on the internet and the method of communication.

C A language developed for creating web pages.

D A system that maps domain names to IP addresses and vice versa.

(2) The following diagram represents the flow of browsing a web page. Choose the term that best fits into the blanks [1] to [3] in the diagram from the options **A** to **D** below, and answer using the letters.



A Web **B** URL **C** IP address **D** DNS

(3) Choose the appropriate combination of names for the blanks [1] to [6] that compose the following URL from the options **A** to **D** below, and answer using the letter.

https://www.mext.go.jp/index.html
[1] [2] [3] [4] [5] [6]

	[1]	[2]	[3]	[4]	[5]	[6]
A	Scheme name	Hostname	Type of organization	Name of organization	Country name	File name
B	Scheme name	Hostname	Name of organization	Type of organization	Country name	File name
C	Hostname	Scheme name	Type of organization	Name of organization	Country name	File name
D	Hostname	File name	Name of organization	Type of organization	Country name	Scheme name

2 Choose the appropriate options for protocols [1] to [3] from the options **A** to **D** below, and answer using the letters.

(1) Protocol for sending emails to a mail server.

(2) Protocol for accessing emails on a mail server.

(3) A protocol that downloads emails from the mail server and deletes them from the server.

A POP **B** SMTP **C** DNS **D** IMAP

8-5 Network Transfer Speed

Point!

Network Transfer Speed

(1) Transfer speed : The ([1]number of bits) of data that can be transferred per second. The unit is ([2]bps).
1 kbps = ([3]1,000 bps), 1 Mbps = ([4] 1,000 kbps), 1 Gbps = ([5]1,000 Mbps)

(2) The relationship between transfer speed, transfer time, and data volume

[1] Transfer speed [bps] = $\left([6]\ \frac{\text{Data volume [bits]}}{\text{Transmission time [seconds]}}\right)$

[2] Transfer time [seconds] = $\left([7]\ \frac{\text{Data volume [bits]}}{\text{Data transfer speed [bps]}}\right)$

[3] Data volume [bits] = ([8]Transfer speed [bps] × Transfer time [seconds])

(3) ([9]Transfer efficiency) : The ratio at which a certain communication speed can be achieved.
Actual transfer speed = ([10]Communication speed [bps] × Transfer efficiency)

Warm Up

Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Calculate the time required to transfer 20 MB of data at a communication speed of 40 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(2) Calculate the time required to transfer 1 GB of data with a communication speed of 100 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.

Explanation

(1) Transfer time [seconds] = $\frac{\text{Data volume [bits]}}{\text{Transfer speed [bps]}}$

1 B = 8 bits, 1 KB = 1,000B, 1 MB = 1,000 KB, therefore,
20 [MB] = 20 × 1,000 × 1,000 × 8 [bits]
Also, 40 [Mbps] = 40 × 1,000 × 1,000 [bps] — Convert units
Therefore, the calculation of transfer time is as follows:

$\frac{20 \times 1{,}000 \times 1{,}000 \times 8\ [\text{bits}]}{40 \times 1{,}000 \times 1{,}000\ [\text{bps}]} = 4\ [\text{seconds}]$ 4 seconds

(2) 1 [GB] = 1 × 1,000 × 1,000 × 1,000 × 8 [bits]
Moreover, since the transfer efficiency is 80%, the actual transfer speed is 100 [Mbps] × 0.8 = 80 [Mbps]
80 [Mbps] = 80 × 1,000 × 1,000 [bps], therefore,
the calculation of transfer time is as follows:

$\frac{1 \times 1{,}000 \times 1{,}000 \times 1{,}000 \times 8\ [\text{bits}]}{80 \times 1{,}000 \times 1{,}000\ [\text{bps}]} = 100\ [\text{seconds}]$ 100 seconds

Try

Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Bps is a unit used to measure data communication speed. Choose the appropriate explanation for 20 Mbps from the options **A** to **D** below, and answer using the letter.
- **A** Capable of transferring data at 20 megabits per minute.
- **B** Capable of transferring data at 20 megabytes per minute.
- **C** Capable of transferring data at 20 megabits per second.
- **D** Capable of transferring data at 20 megabytes per second.

(2) Calculate the maximum data volume, in bytes, that can be transmitted in one minute at a communication speed of 30 bps.

(3) Calculate the maximum data volume, in MB, that can be transmitted in five minutes at a communication speed of 10 Mbps.

(4) Calculate the time required to transfer 20 MB of data at a communication speed of 80 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(5) Calculate the time required to transfer 1 GB of data at a communication speed of 20 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.

Exercise

1 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, 1 MB = 1,000 KB, and 1 GB = 1,000 MB.

(1) Bps is a unit used to measure data communication speed. Choose the appropriate explanation for 40 Mbps from the options **A** to **D** below, and answer using the letter.
- **A** Capable of transferring data at 40 megabits per second.
- **B** Capable of transferring data at 40 megabytes per second.
- **C** Capable of transferring data at 40 megabits per minute.
- **D** Capable of transferring data at 40 megabytes per minute.

(2) Calculate the maximum data volume, in bytes, that can be transmitted in one minute at a communication speed of 120 bps?

(3) Calculate the data volume, in MB, that can be transmitted in three minutes at a communication speed of 20 Mbps.

(4) Calculate the time required to transfer 15 MB of data at a communication speed of 60 Mbps. Assume that the transfer efficiency is 100% and disregard any data other than the data volume.

(5) Calculate the time required to transfer 1 GB of data at a communication speed of 25 Mbps. Assume that the transfer efficiency is 80% and disregard any data other than the data volume.